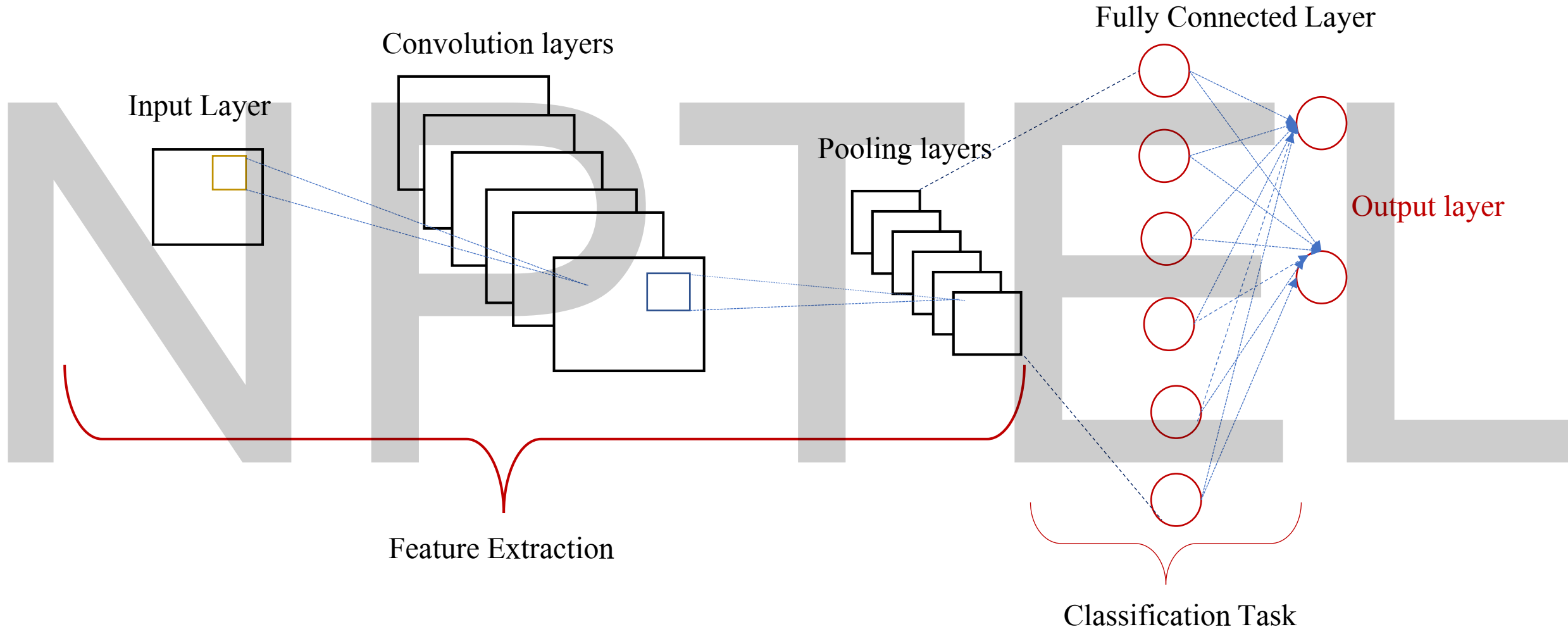


Introduction to Standard CNN Models

Content

NPTEL
Architecture
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Convolution Neural Network Structure



Standard CNN models are the architectures which are known for their application in image recognition problem

Context Before AlexNet

Before 2012, **computer vision** relied heavily on **hand-crafted features**

Handcrafted Features

Manually define a set of features and extract them.

Examples: edge detection, corner detection, histograms, etc.

Handcrafted features **rely heavily on human intuition**: the designer must decide which features are important.

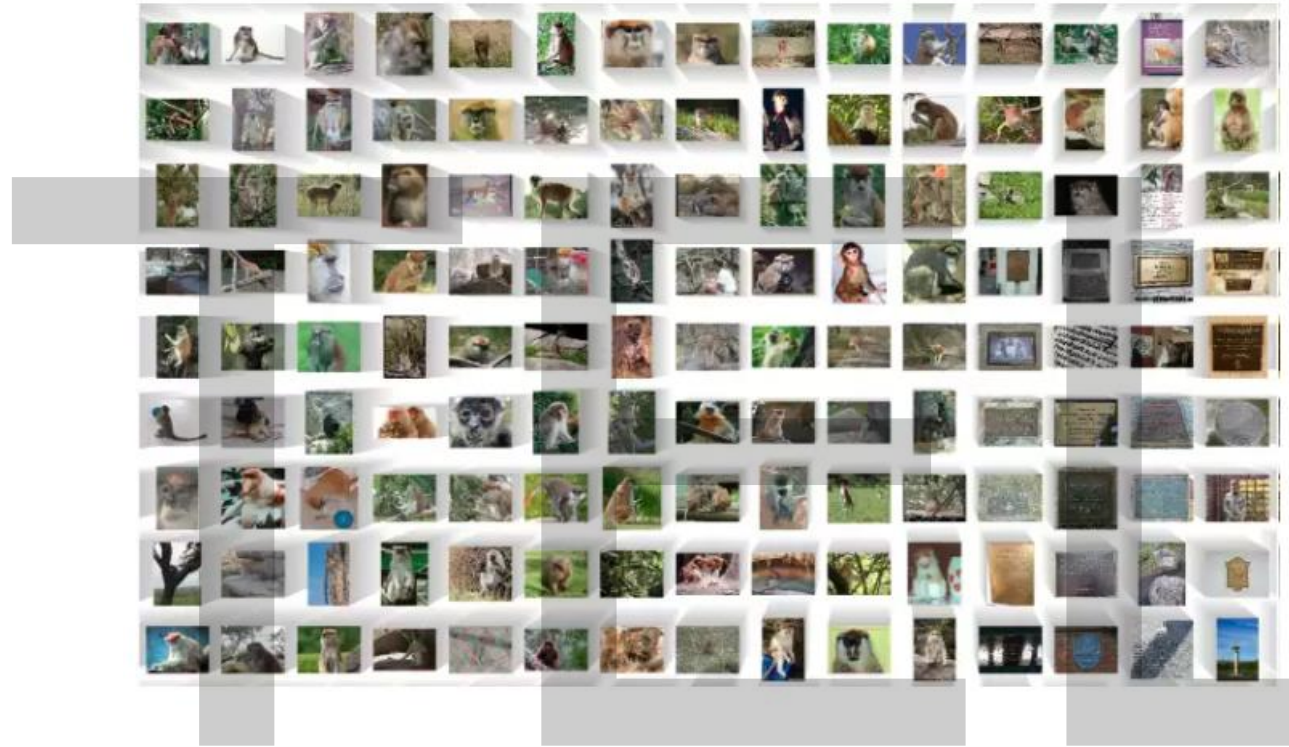
Limitations of handcrafted feature extraction

- ✓ Works well with simple tasks but struggled with large, complex datasets due to
 - Too many variations (different lighting, poses, backgrounds).
 - Can't capture higher-level patterns (patterns of cat class or dog class)

ImageNet Dataset

1.2 million high-resolution images

- Created in 2009 (Fei-Fei Li, Stanford University)
- 1000 categories (synsets from WordNet)
- Benchmark dataset for object recognition



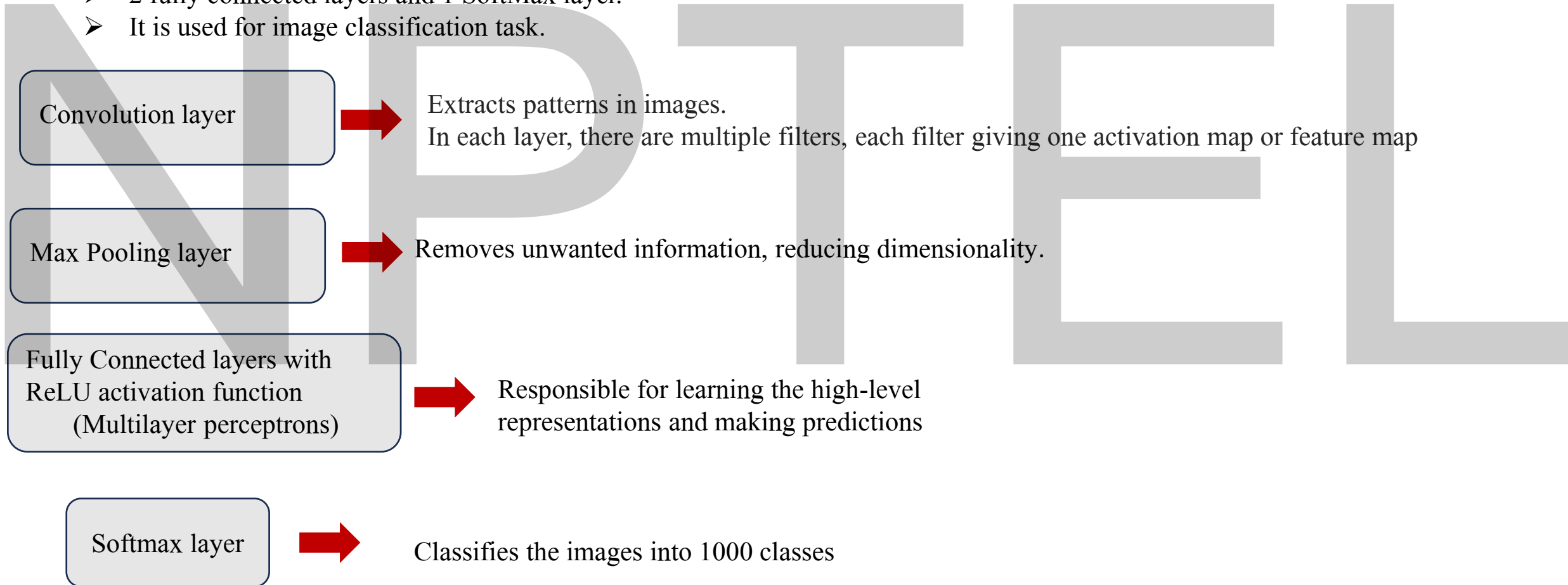
ImageNet dataset(2009)

- ✓ WordNet® is a large lexical database of English.
- ✓ Synsets are groups of cognitive synonyms of Nouns, verbs, adjectives and adverbs
- ✓ Each synset expresses a distinct concept

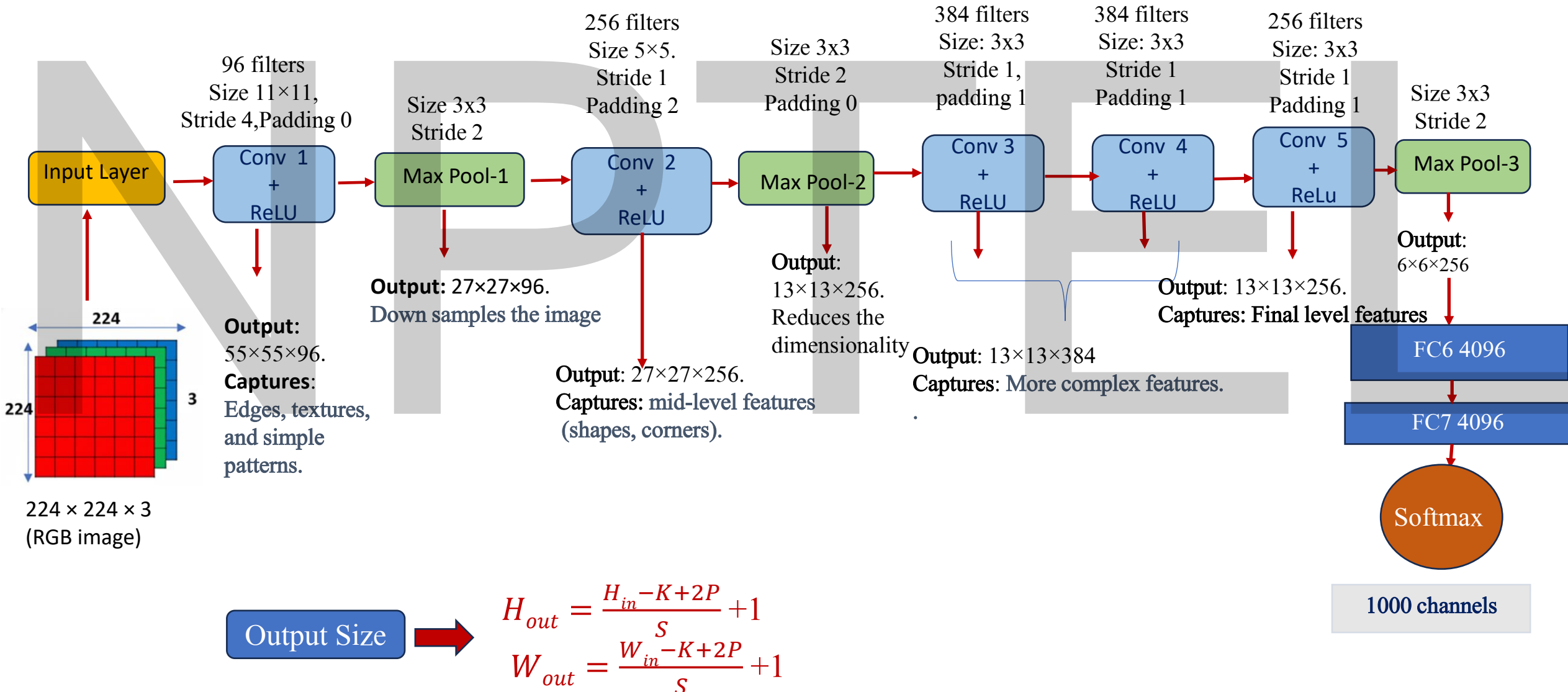
- ImageNet Large Scale Visual Recognition Challenge (ILSVRC) was a benchmark competition in computer vision that ran annually from 2010 to 2017.
- It was designed to measure the progress of algorithms in recognizing and understanding objects in images on a large scale.
- The best models at that time could barely reach **74% top-5 accuracy** (which means they made a mistake in 26% of the top-5 guesses).
- **Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton** from the University of Toronto trained a **deep convolutional neural network (CNN)** using GPUs on ImageNet.
- They named it as **AlexNet**.

Introduction to AlexNet (2012)

- AlexNet is a **deep Convolutional Neural Network (CNN)** consisting of
 - 5 convolution layers,
 - 3 max-pooling layers,
 - 2 fully connected layers and 1 SoftMax layer.
 - It is used for image classification task.



AlexNet Architecture



Types and Characteristics of Image Features

Low-level Features

Low-level image features include color, texture, shape, edges, and corners, which are extracted directly from pixel values or small groups of pixels in an image.

Mid-Level Features

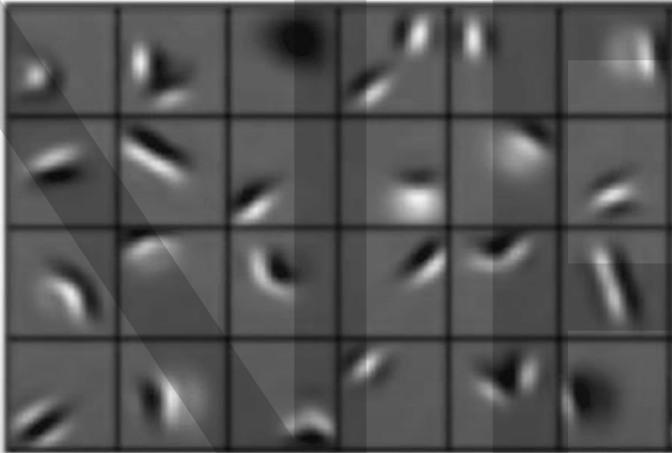
These describe **larger patterns or structures** that emerge from combinations of low-level features.
They often represent **parts, shapes,**
They more meaningful than pixels, but not yet a whole object.

High level Features

These correspond to **object-level representations**

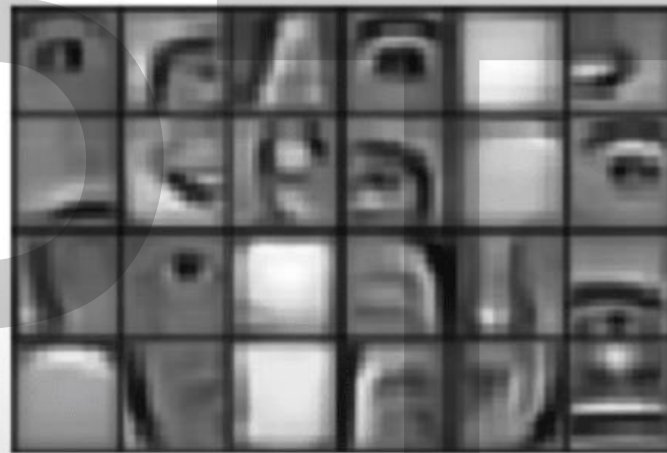
Example of features in an images

Low level features



Edges, dark spots

Mid level features



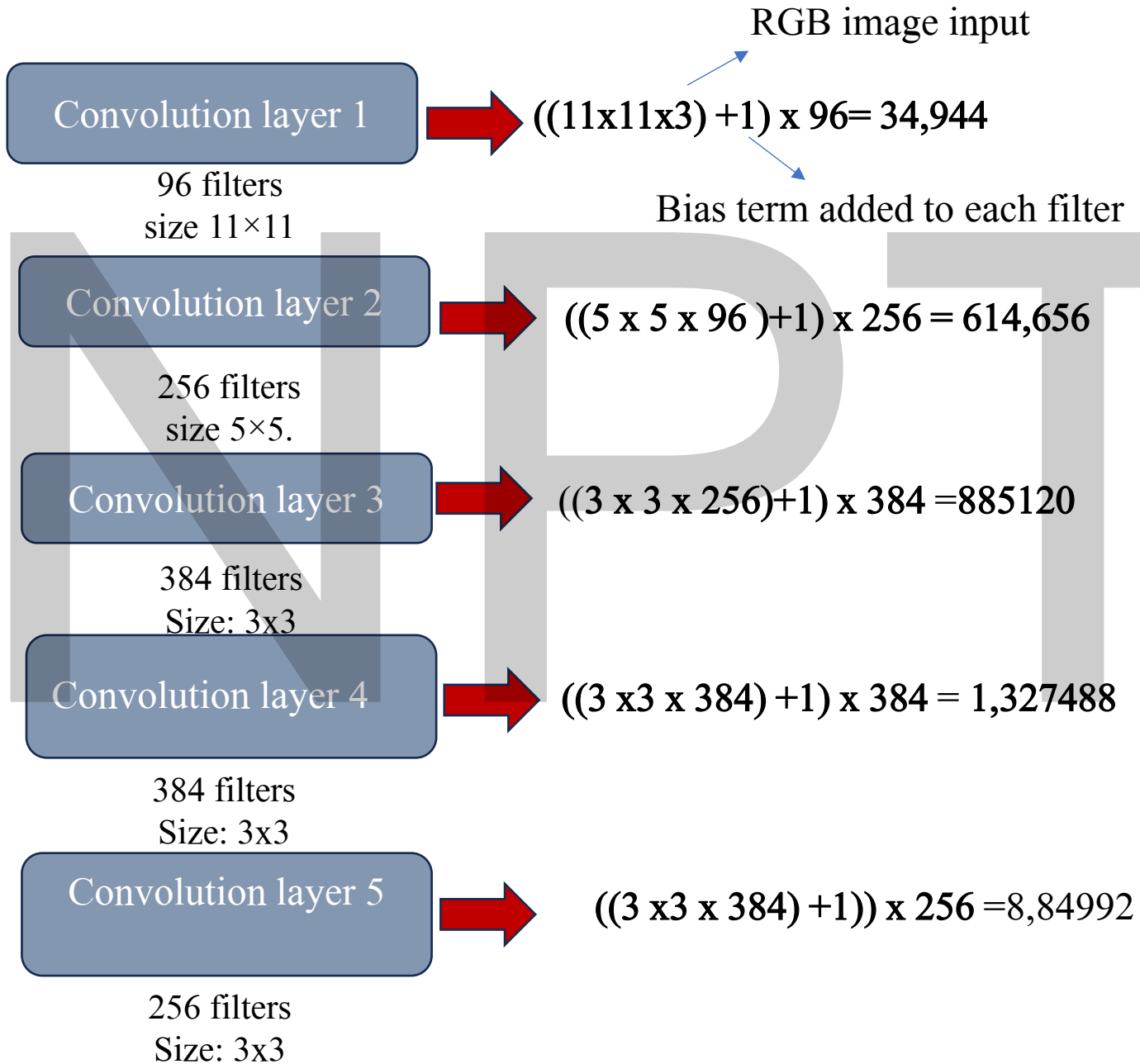
Eyes, ears, nose

High level features



Facial structure

Parameter calculation for each Convolution layer

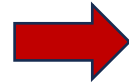


Layers	Parameters
Convolution layer 1	35 K
Max pool 1	No learnable parameter
Convolution layer2	614 K
Max pool 2	No learnable parameter
Convolution layer 3	885K
Convolution layer 4	1.32 M
Convolution layer 5	8.84 M
Max pool 3	No learnable parameter

Parameter calculation for the fully connected layers

Fully connected layer 1

Input size: 9216
Nodes: 4096



$$(9216 \times 4096) + 4096 = 37,752,832$$

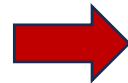
Fully connected layer 2

Nodes: 4096



$$(4096 \times 4096) + 4096 = 16,781,312$$

Softmax layer



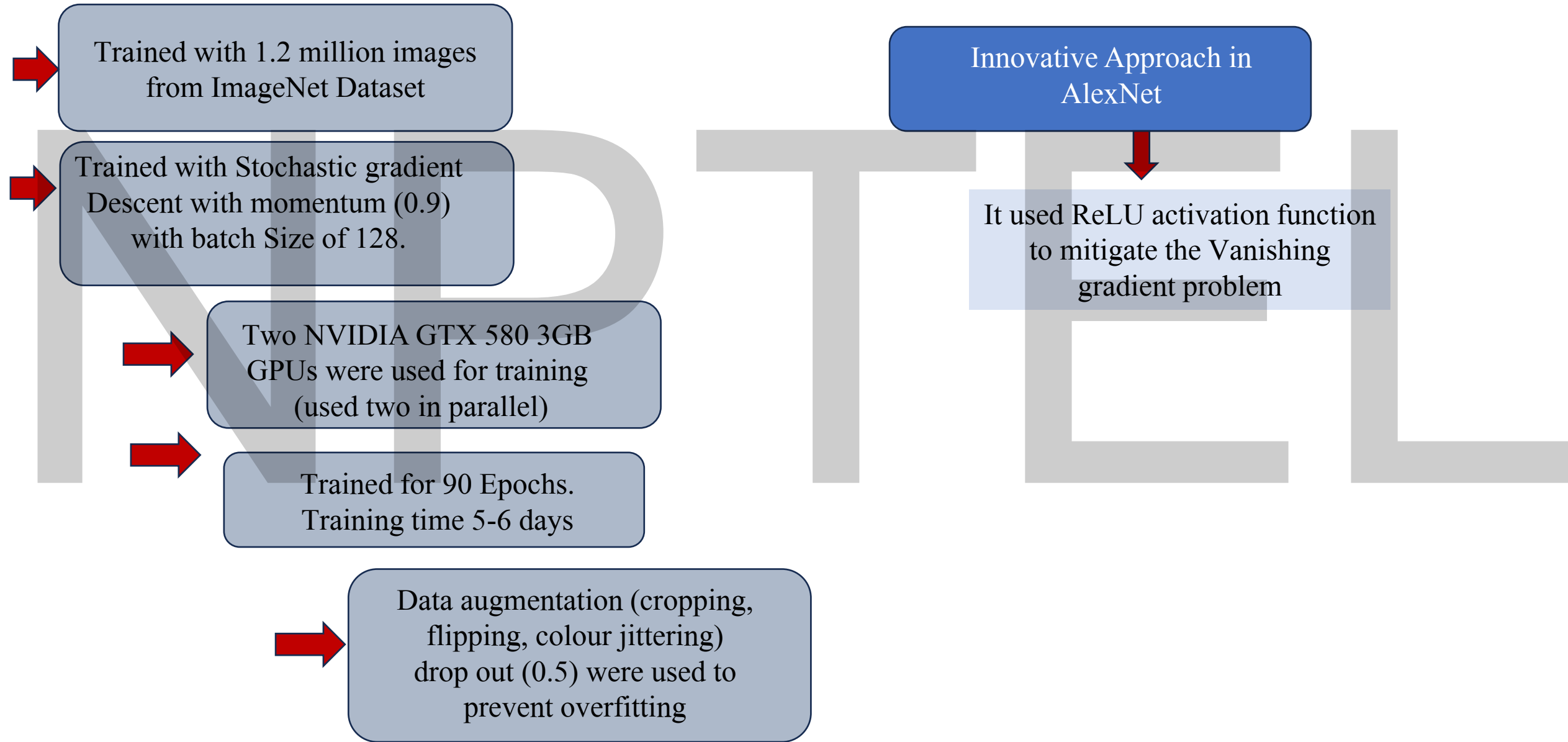
$$(4096 \times 1000) + 1000 = 4,097,000$$

No of Channels=1000

Layers	Parameters
Fully connected layer 1	37 M
Fully connected layer 2	16 M
Soft max layer	4 M

Total no of parameters in Alexnet : 62.4 Million
95% of the parameters are coming from the fully connected layers

Implementation of AlexNet



ImageNet Large Scale Visual Recognition Challenge-2012

- **In ILSVRC 2012, AlexNet achieved a top-5 error rate of 15.3%**
- **It is more than 10 percentage points better than the next best competitor at 26.2%.**
- This is a dramatic improvement over prior models which relied on shallow learning and hand-engineered features

The success of Alexnet inspired models like VGGNet, GoogleNet and ResNet

NPTEL

Summary

NPTTEL

Thank you